Assignment 1, Matrix Theory

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Exercise 1

Proof. Let $x, y \in \mathcal{C}$ and $z = \lambda x + (1-\lambda)y$ for some $\lambda \in (0,1)$.

($\Rightarrow$) Assume $a_i^T z = b_i$. By linearity,

$$a_i^T z = a_i^T(\lambda x + (1-\lambda)y) = \lambda\, a_i^T x + (1-\lambda)\, a_i^T y = b_i.$$

Since $x, y \in \mathcal{C}$, $a_i^T x \leq b_i$ and $a_i^T y \leq b_i$. Suppose for contradiction that $a_i^T x < b_i$. Then, using $\lambda, (1-\lambda) \in (0,1)$,

$$\lambda\, a_i^T x + (1-\lambda)\, a_i^T y < \lambda\, b_i + (1-\lambda)\, b_i = b_i,$$

contradicting $a_i^T z = b_i$. Hence $a_i^T x = b_i$, with the same reasoning $a_i^T y = b_i$.

($\Leftarrow$) Assume $a_i^T x = b_i$ and $a_i^T y = b_i$. Then $a_i^T z = \lambda\, a_i^T x + (1-\lambda)\, a_i^T y = \lambda\, b_i + (1-\lambda)\, b_i = b_i$. □